



भारतीय सूचना प्रौद्योगिकी संस्थान गुवाहाटी
Indian Institute of Information Technology Guwahati

CS236 - ARTIFICIAL INTELLIGENCE LAB
ASSIGNMENT - 03

Part 1: Pandas

1. Create a Pandas DataFrame named `df` with columns `Std_Name` (string), `Roll_no` (integer), and `CPI` (float). Write Python code to create a new DataFrame containing only the rows where the value in the `CPI` column is greater than 60. Also, calculate the overall mean, median, and standard deviation of the `CPI` column.
2. Suppose you have a Pandas DataFrame named `IIITG_Library` with columns `Subjects` (string), `Book_Authors` (string), and `No_of_Books` (integer). Write Python code to calculate the total number of books for each unique subject across all authors.
3. Write a Python program to read an image file (e.g., .png, .jpg) and save the image pixel data as a numerical matrix (NumPy array) to a .csv file using Pandas. Then, demonstrate how to import the numerical data from the .csv file back into a Pandas DataFrame, excluding the last row and the last column of the .csv file.
4. Suppose you have a Pandas DataFrame named `dirty_data` with columns `Name` (string), `Age` (integer), and `Salary` (float). Write Python code to remove rows where the `Age` is less than 18 or the `Salary` is negative.
5. Given a Pandas DataFrame named `time_df` with a column `TimeStamp` containing datetime objects (e.g., `pd.to_datetime` format), write Python code to create a new column named `Hour` containing the hour component of each timestamp in the `TimeStamp` column.

Part 2: NumPy Application

1. Create two NumPy matrices, P and Q , each of size 106×104 filled with random floating-point values. Perform the following, measuring the execution time for each step:
 - i. Calculate the matrix multiplication $P \cdot Q^T$ using nested loops in Python. Store the execution time as t_1 .
 - ii. Calculate the matrix multiplication $P \cdot Q^T$ using NumPy's vectorized operations (e.g., `@` or `numpy.dot`). Store the execution time as t_2 .
 - iii. Calculate and report the speedup as $\text{Speedup} = t_1/t_2$.
2. Given a 2D NumPy array named `arr`, write Python code to extract the second column and the last row as separate NumPy arrays.

3. Create a 2D NumPy array of shape 5×6 containing random integers. Write Python code to find the frequency of each unique number in the array.
4. Solve the system of linear equations $Ax = b$ using NumPy, where A is a randomly generated $n \times n$ coefficient matrix, x is an $n \times 1$ unknown vector, and b is a randomly generated $n \times 1$ vector. Choose a specific value for n (e.g., $n = 5$).
5. In a mathematics class, students took a quiz, and their scores are recorded as follows:

85, 92, 75, 85, 90, 92, 85, 75, 85, 92, 75, 85, 90, 92, 85, 75, 85, 92.

Write Python code using NumPy to calculate the frequency of the score 85.

6. Calculate the eigenvalues and eigenvectors of the following matrix A using NumPy's `linalg` module:

$$A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$$

7. Given a matrix X representing a dataset where rows are data points and columns are features:

$$X = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

Standardize the features (columns) of X by subtracting the mean and dividing by the standard deviation of each column. Use NumPy's broadcasting capabilities to accomplish this efficiently.

8. Perform Singular Value Decomposition (SVD) on the following matrix M using NumPy's `linalg` module:

$$M = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$$

Part 3: Data Visualization with Matplotlib/Seaborn

1. **Book Inventory Visualization:** You are given the following data representing a simplified inventory of a library:

Subject	Author	Quantity
Mathematics	R.D. Sharma	15
Physics	H.C. Verma	12
Chemistry	O.P. Tandon	8
Mathematics	S.K. Goyal	10
Physics	Resnick	5

Create a bar chart using either Matplotlib or Seaborn to visualize the total number of books available for each subject. Ensure the following:

- The chart has a title: "Total Books per Subject".

- The x -axis is labeled "Subject".
- The y -axis is labeled "Number of Books".
- The bars are clearly labeled or distinguishable.

2. **Salary Distribution Visualization:** Consider the following dataset representing the salaries (in thousands of dollars) of employees in a small company:

55, 62, 65, 58, 70, 55, 60, 68, 63, 55, 62, 150

Create a histogram using Matplotlib or Seaborn to visualize the distribution of these salaries. Choose an appropriate number of bins to represent the data distribution. Ensure that:

- The chart has a title: "Distribution of Employee Salaries".
- The x -axis is labeled "Salary (in thousands of dollars)".
- The y -axis is labeled "Frequency".